

OH-XTRA®

THE SUPERIOR SURFACE.

THE QUALITY OF THE PREFORMS, ALONG WITH OTHER ASPECTS, IS DECISIVE FOR THE EFFICIENCY OF THE DOWNSTREAM PRODUCTION. HIGHLY INFLUENTIAL FACTORS HERE ARE THE PROPERTIES OF MOULDS AND ESPECIALLY CORE SURFACES. WITH OH-XTRA®, OTTO HOFSTETTER AG HAS DEVELOPED A NEW TECHNOLOGY WHICH IS WAY AHEAD OF ALL CONVENTIONAL SURFACE TREATMENTS OF MOULD CORES.

The core of quality.

In the case of a PET injection mould, the surfaces exposed to the processed plastic material are decisive in many respects because they eventually determine not only the shape, but also a big part of the quality of a preform. On the one hand, we strive



Cores OH-Xtra®.

to achieve a perfect surface structure on the steel parts to provide the actual preform with a regular, smooth surface. On the other, a structure in the direction of ejection is an essential advantage for onward processing, so that ejection from the mould takes place very smoothly and the plastic part is protected from any deformation. Here, particular attention is paid to the cores. During production – as we know – the preform remains in contact with this mould component for the longest part of the cycle, and is also steadily cooled there.

The search for the ideal surface.

Looking at the development history of PET moulds, it is clear that many different surface treatments and coatings have already been applied to steel cores for

PET: high-gloss polish, chrome-plated surface, the golden TiN coating and many more. The objectives of these surface treatments and coatings were and still are always to attain the above-described smoothest possible surface that allows to eject the preforms without resistance.

Cycle time and material are decisive too.

The specified cycle times and the materials to be processed also have a major influence on the quality of the preforms. Both factors must be taken up accordingly in the overall consideration of the preform quality. While the marked pressure of costs leads to ever shorter cycle times, the wish for a more ecological production causes the portion of recycled material to rise considerably. The two aspects must be taken into account but are not allowed to deteriorate the preforms' quality properties.

OH-Xtra® sets new benchmarks.

The developers of Otto Hofstetter AG have met this challenge and set out to develop an entirely new surface treatment for the mould cores. Through intensive research they discovered a method which appeared to satisfy all requirements. In the subsequent trials and test series, the newly developed technique confirmed the predictions of the experts.

OH-Xtra® is the name of the new surface treatment for cores. The method is applied directly to the mould part using a special process which results in striking advantages for both the preforms and the mould:

- No micro-scratches on the preforms any more, especially in the mouth section in direction of ejection
- Less deposits of dirt particles on the actual moulding surfaces (lotus flower effect) and thereby a reduced cleaning effort
- Lower ejection forces are required in the mould

OH-Xtra® has been used in all moulds manufactured by Otto Hofstetter AG since January 2017. Cores of earlier generations of Hofstetter moulds can be retrofitted with OH-Xtra® without restriction, of course.

